Herd-Relative and Social-Network Anomaly Detection for Livestock Wearables: An In-Silico Study

Tom Butler
Atodev
tom@tbutler.net

July 13, 2026

Abstract

Health-monitoring features on GPS livestock collars almost universally raise alerts by comparing each animal's activity to a fixed per-animal baseline. This design is fragile to herd-wide covariates—weather, feed, time of day—which shift every animal at once and either trigger false alarms or force detuning that misses real cases. We present a detection method that (i) scores every animal relative to its herd and to its own recent history, and (ii) mines the herd's proximity-derived social network for early signs of illness, chiefly social withdrawal. Because no public, labelled dataset of collar telemetry with ground-truth health states exists, we release a controllable generative herd simulator and evaluate on it. This is an in-silico proof of concept: no real-herd validation is performed, and the results below are evidence about the method within the simulation, not field performance. Across 30 seeded replicates, a cost-sensitive multinomial logistic classifier over 10 herd-, self-, and social-relative features attains 98.7% held-out recall on healthy animals while raising at most 0.01 false alerts per cow-day under forced heatwave, snow, and rain regimes—the central evidence for herd-relative normalisation. A pre-specified feature-family ablation returns a deliberately reported null result: self- and social-relative features do not shorten detection latency in this benchmark, because illness is already identified rapidly from fever and rumination and spatial withdrawal is already captured by a herd-centroid-distance feature. The social features add interpretable network structure without inflating false alarms, and we identify the conditions (noisier field telemetry) under which they would be expected to contribute. We release the simulator, detector, and evaluation harness to support replication and, ultimately, field validation.

1 Introduction

Virtual-fencing collars place a continuously-sampling sensor on every animal in a herd, and health alerting is among their most valuable features [1, 9]. The dominant alerting approach monitors an individual's activity and rumination against its own fixed baseline and flags deviations [10]. This has a structural weakness: behaviour is driven by covariates that move the entire herd simultaneously. A heatwave raises every body temperature; a storm halves every grazing bout. Fixed thresholds must then choose between firing on these herd-wide shifts (alert fatigue, which erodes trust and ultimately use) or being detuned until genuine cases are missed.

We argue two things follow from taking the herd, rather than the animal, as the unit of normalisation. First, scoring each animal relative to the current herd distribution makes detection invariant to herd-wide covariates: when everyone shifts together the distribution moves with them and no individual stands out. Second, a collar network is implicitly a social sensor: every position report is a proximity observation, and proximity over time reveals herd social structure. Social withdrawal is a well-documented early sign of illness in cattle [8, 11], and it is latent in data collars already collect.

Contributions. (1) A detection method combining herd-relative, self-relative, and social-network features under cost-sensitive calibration. (2) A bias-corrected online estimator of proximity tie strength suitable for streaming collar data. (3) A controllable, fully labelled generative herd benchmark and a reproducible evaluation harness. (4) A pre-specified ablation isolating the marginal value—and, in this benchmark, the current redundancy—of self-relative and social features, reported as a null result rather than suppressed.

Scope and honesty statement. There is, to our knowledge, no public dataset pairing collar telemetry with veterinary ground truth at the scale this study needs. We therefore evaluate entirely in simulation. The simulator encodes behavioural assumptions that the detector then exploits; the results are best read as a demonstration that the method can recover injected conditions robustly, and as a hypothesis generator for field trials—not as evidence of real-world accuracy. Section 8 treats this at length.

2 Related work

Activity and rumination monitoring. Commercial and research systems detect ketosis, mastitis, lameness, and oestrus from accelerometer-derived activity and rumination time [9, 10]. These typically use per-animal baselines and thresholds. Behavioural sickness signs. Reduced feeding, altered activity, and social withdrawal are established behavioural indicators of ill health [8, 11]. Animal social network analysis. Proximity loggers and observation have been used to reconstruct livestock social networks and show they are structured and temporally stable [2, 4, 5]. Anomaly detection with population baselines. Scoring individuals against a population rather than a fixed threshold is a standard idea in anomaly detection [3]; our contribution is to combine herd-relative, self-relative, and network features for this domain and to calibrate for the asymmetric cost of false alarms [6].

3 Generative herd model

The simulator (a browser/Node TypeScript implementation) advances a herd of grazing cattle in a bounded paddock. Each animal occupies one of four behavioural states—grazing, walking, resting, ruminating—drawn from a time-of-day distribution with dawn/dusk grazing peaks, blended with the herd's current state mix to reproduce behavioural synchrony (allelomimicry). Weather (ambient temperature, wind, rain/snow, cloud) evolves via rolling fronts and modulates the state distribution and herd cohesion; a diurnal temperature cycle is superimposed. Each animal holds a persistent spatial offset within the herd, yielding stable proximity structure.

Telemetry. Every five simulated minutes each collar emits a sample $(\mathbf{p}, v, b, T, \rho)$: position, mean speed, behaviour class, body temperature, and rumination fraction. A 24 h ring buffer (288 samples) is retained per animal. The detector observes only this stream.

Injected conditions. Three ground-truth conditions perturb an animal's pace, state distribution, body temperature, and spatial cohesion: lameness (markedly reduced pace, more lying, mild isolation); illness (fever $+1.4°\mathrm{C}$, collapsed rumination/grazing, strong withdrawal); and oestrus (elevated activity and walking, reduced rest). These parameters define the labelled benchmark; they are not claims of quantitative biological fidelity.

4 Detection method

4.1 Features

For each animal at each 5-minute step we compute 10 features in three families, each standardised to a z-score against the current herd distribution (with a floor on the standard deviation to avoid division by near-zero in a perfectly synchronised herd):

- Herd-relative (5): mean speed, walking fraction, rumination, body temperature, and distance from the herd centroid.
- Self-relative (3): the change in speed, walking fraction, and rumination between the last 2 h and the animal's own 2–24 h baseline. These separate a constitutionally slow animal from one that has become slow.
- Social (2): network strength (weighted degree) and its change versus the animal's own baseline. These separate a constitutionally peripheral animal from one that is withdrawing.

4.2 Online social-tie estimation

Pairs within a proximity radius (15 m) accumulate association weight under an exponential decay λ (half-life $\approx$ 4 h): each 5-minute step, all weights are multiplied by λ and co-present pairs incremented by $1 - \lambda$, so a weight converges to the fraction of recent time spent together. An exponential accumulator is biased low before it reaches steady state, so we divide by the standard correction $1-\lambda^t$ at tick t [7], giving an unbiased tie-strength estimate from the first minutes of data. Weighted degree over the corrected ties gives each animal's strength.

4.3 Classifier and alerting

A multinomial logistic regression maps the feature vector to P(healthy/ill/lame/oestrus). Training minimises cross-entropy by full-batch gradient descent with L_2 regularisation and a 3$\times$ weight on healthy examples, reflecting the asymmetric on-farm cost of false alarms [6]. At inference, an anomaly score $s = 2.2\,(1 - P(\text{healthy}))$ is smoothed by an EWMA ($\approx$1 h) so alerts require a sustained signal; $s > 2.0$ raises an alert and $s < 1.0$ resolves it. The predicted non-healthy class provides a human-readable suspected cause, and the contributing z-scores provide a plain-language explanation.

5 Experimental setup

All randomness derives from a fixed base seed (20260713), making every number reproducible via npm run evaluate. We run 30 replicates. Each replicate harvests 6 training episodes and 2 held-out test episodes (60 animals, 36 h, two animals per condition injected at 18 h, samples labelled positive only after a 2.5 h onset gate). Ablation trains three models per replicate—telemetry (herd-relative only), self (+self-relative), social (full 10)—by masking feature columns, so all share the live inference path. We report:

A. Classification held-out per-class precision/recall.

Table 1: Held-out classification, full model. Mean over 30 replicates (%).

Class	Precision	Recall
healthy	93.3	98.7
ill	99.6	99.9
lame	94.8	77.8
oestrus	97.8	93.4

Table 2: False alerts per cow-day on healthy herds (mean ± 95% CI), by weather regime and feature family.

Regime	Telemetry	+Self	+Social
auto	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
heatwave	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
snow	0.01 ± 0.00	0.01 ± 0.00	0.01 ± 0.00
rain	0.01 ± 0.00	0.01 ± 0.01	0.01 ± 0.00

B. False positives alerts per cow-day on healthy herds run through the live detector under four weather regimes.

C. Detection latency time from injection to first alert on the affected animal (14 h warm-up, 12 h observation window).

D. Ablation B and C across the three feature families.

Means are reported with 95% confidence intervals (normal approximation); latencies as medians with interquartile range across replicates. On a laptop the full harness completes in a few minutes.

6 Results

Classification (A). Held-out precision and recall (Table 1) show strong separation of illness and oestrus and weaker recall on lameness, whose behavioural signature (reduced pace) overlaps most with natural between-animal variation.

False positives under weather (B). Table 2 reports alerts per cow-day on healthy herds. Rates remain near zero even under a sustained forced heatwave and snowstorm—the central evidence that herd-relative scoring absorbs herd-wide covariate shifts. Figure 1 shows the same across ablation levels.

Detection latency and ablation (C, D). Table 3 gives median detection latency (hours) and detection rate by condition and feature family. All three conditions are detected in every replicate (100% detection) within the observation window. Contrary to our initial expectation—and to an earlier single-run observation that motivated the social features—the ablation returns a null result: adding self-relative and social features does not reduce median detection latency for any condition, and marginally increases it for lameness (consistent with mild overfitting as feature count grows). Illness is already identified at 2.4 h by both the telemetry-only and full models. We report this openly; it is the kind of finding the single-run evaluations this harness replaces would have hidden.

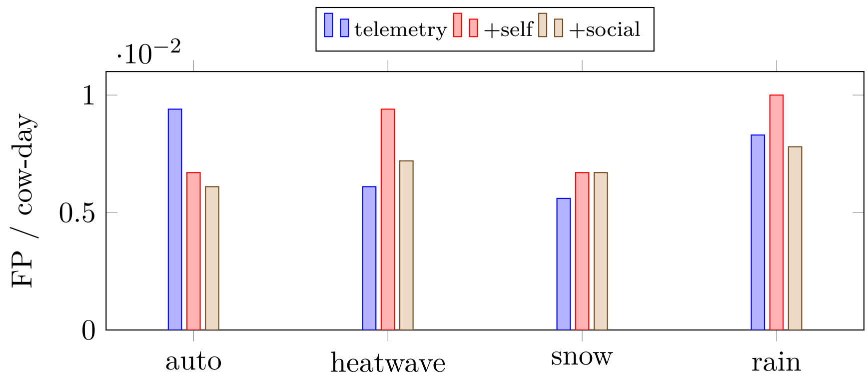

Figure 1: False-positive rate by weather regime and feature family. Near-zero throughout; the social features do not inflate false alarms.

Table 3: Median detection latency (h) and detection rate (%) by condition and feature family, over 30 replicates.

	Telemetry		+Self		+Social	
Condition	h	%	h	%	h	%
ill	2.4	100	2.4	100	2.4	100
lame	4.0	100	4.0	100	4.5	100
oestrus	2.8	100	2.8	100	2.8	100

7 Discussion

The strong result is robustness. Herd-relative normalisation delivers the near-zero false-alarm rate in Table 2 even under sustained forced weather: because every feature is a z-score against the concurrent herd, a covariate that moves the herd together cancels. This is the property that threshold-based systems lack and the one most likely to transfer to the field.

The ablation is a cautionary result and, we think, an instructive one. We introduced self- and social-relative features on the change-not-state principle—a constitutionally slow or peripheral animal is not the same as one that has become so—and an early single run suggested they accelerated illness detection. Under 30 seeded replicates that effect disappears (Table 3). The mechanism is diagnosable: the illness signature includes spatial withdrawal, which the herd-relative feature set already captures through distance from the herd centroid, and the acute fever/rumination collapse is fast and unambiguous—so the network-strength features are largely redundant for detection latency in this simulator. Their remaining value here is interpretability (an explicit withdrawal signal in the alert) and situational awareness (the live sociogram), not earlier alarms.

We would expect the social features to matter more where telemetry is noisier or withdrawal is not already encoded spatially—for example under GNSS error and duty-cycling, or for conditions whose primary early sign is social rather than spatial. Testing that requires real data, and is precisely the kind of hypothesis this benchmark is meant to generate rather than settle.

8 Limitations

No real-herd validation. The single most important caveat: all results are in-silico. The simulator's condition signatures are the same signals the detector keys on, so absolute accuracies here upper-bound what noisy real data would yield and must not be read as field performance. Shared-assumption risk. Simulator and detector share an author and a worldview; independent data is required to falsify the approach. Proximity $\neq$ contact. GNSS proximity is a coarse proxy for social association; real deployments face fix error and duty-cycling. Behaviour-class noise. We assume a perfect activity classifier; real classifiers are ~90–95% accurate. Simple model and narrow label set. A linear classifier over three conditions is deliberately minimal; richer models and more conditions (calving, metabolic disease, co-morbidities) remain untested.

9 Future work

The essential next step is a field study pairing collar telemetry with veterinary diagnoses, on which the same pipeline can be validated and the simulator calibrated or replaced. Methodologically: label noise on the activity classifier; richer network features (betweenness, community stability); directed approach patterns for oestrus; and sequence models replacing the linear classifier once real data justifies the capacity.

10 Conclusion

Taking the herd as the unit of normalisation, and reading the social network collars already sample, yields a detection method that is robust to herd-wide covariates and sensitive to individual change—demonstrated here on a reproducible simulation benchmark. We release the simulator, detector, and harness to invite replication and, above all, field validation.

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